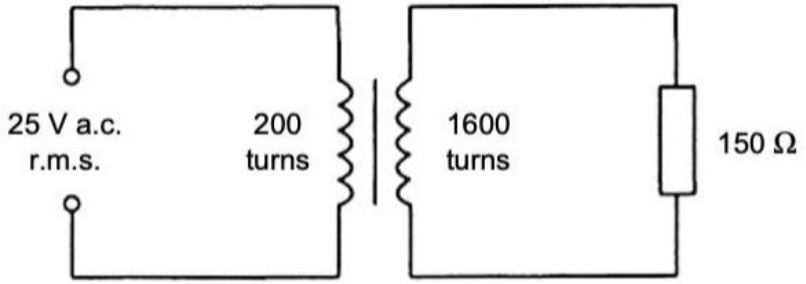
21 The primary of an ideal transformer has 200 turns and is connected to a 25 V r.m.s. alternating current supply. The secondary has 1600 turns and is connected to a resistor of resistance 150 Ω, as shown below.



What are the possible values of the secondary voltage, the secondary current and the mean power dissipated in the resistor?

	secondary voltage / V r.m.s.	secondary current / A r.m.s.	resistor power / W
A	3.1	0.021	0.065
B	25	0.17	4.2
C	200	1.3	270

D	200	1.3	530

22 A capacitor is charged by a constant current of 2.5 μA for 200 s. As a result, the potential difference